ZeroBleed results report

Generated by `npm run report` from the raw files in `evidence/`. Every number below is computed from those files; nothing is typed by hand.

How to check these results are real

1. **Model calls.** Each run has a `*.calls.json` file listing every model call with the model name, the token usage Nightshift reported, and the Nightshift **receipt ID**. Open Nightshift's **Receipts** view and look up any receipt ID: it shows the same request, model and token counts. Our code cannot write to that view.
2. **Data.** Each fact a decision used is listed with the exact Zetaris SQL that produced it (`evidence` in `*.calls.json`). Run the same SQL in the Zetaris SQL Editor at http://localhost:3000 and you get the same value.
3. **Steps.** Each run has a `*.trace.jsonl` file: one line per planner, worker, gate, guard, Markovian chunk and decision, with timestamps and models.
4. **Reproduce.** `npm run benchmark -- procurement 1` and `npm run benchmark -- incident 1`, then `npm run report`. Models can vary slightly between runs; the correct answers are fixed in `src/domains/*.ts`.

Failures stay in the denominator. Token counts are input tokens as reported by Nightshift, which include Nightshift's own per-request overhead for both systems alike.

Headline (current runs)

Domain	Runs	Correct: ZeroBleed	Correct: shared-history agent	Input tokens: ZeroBleed	Input tokens: shared-history	Fewer tokens	Largest prompt: ZeroBleed vs shared
Supplier penalty audit	4	**4 / 4**	3 / 4	17,133	197,152	**91.3%**	880 vs 10,073
Incident triage	4	**4 / 4**	1 / 4	28,860	726,511	**96.0%**	1,995 vs 63,860

Supplier penalty audit

Measured 2026-10-23 on MacBook Air (Apple M5, 16 GB), Zetaris 2.4.3.1 in Docker, Meterless Nightshift Mac M5+. Fast lane `nvidia/nemotron-3.5-lightning-30b-a3b`, deep lane `nvidia/nemotron-3-ultra-550b-a55b`. 4 runs.

	ZeroBleed	Shared-history agent
Correct decisions	**4 / 4**	3 / 4
Input tokens, total	**17,133**	197,152
Largest single prompt	**880**	10,073
Median wall time per case	23.45s	13.7s

ZeroBleed read **91.3% fewer input tokens**.

Run	Scenario	Expected	ZeroBleed	Shared-history agent	Evidence
SHP-1017-r1	21 days late, no waiver	ENFORCE 48000	✅ ENFORCE 48000 · 4,492 tok	✅ ENFORCE 48000 · 49,442 tok	calls · trace
SHP-1029-r1	18 days late, storm waiver active	WAIVE 0	✅ WAIVE 0 · 3,368 tok	❌ ENFORCE 30000 · 49,506 tok	calls · trace
SHP-1036-r1	9 days late, under the 14-day threshold	NO_BREACH 0	✅ NO_BREACH 0 · 4,223 tok	✅ NO_BREACH 0 · 49,069 tok	calls · trace
SHP-1044-r1	25 days late, contract text carries a prompt injection	ENFORCE 38400	✅ ENFORCE 38400 · 5,050 tok	✅ ENFORCE 38400 · 49,135 tok	calls · trace

Receipts from `SHP-1017-r1.calls.json` (look these up in Nightshift > Receipts):

Step	Model	Input tokens	Nightshift receipt
ZeroBleed PLAN	nvidia/nemotron-3-ultra-550b-a55b	692	`rcpt_req_pdmd6x0h`

Step	Model	Input tokens	Nightshift receipt
ZeroBleed T1_EXTRACT	nvidia/nemotron-3.5-lightning-30b-a3b	880	`rcpt_req_b83083md`
ZeroBleed T2_CRITIQUE	nvidia/nemotron-3.5-lightning-30b-a3b	709	`rcpt_req_6eyks07j`
ZeroBleed T2_CRITIQUE	nvidia/nemotron-3.5-lightning-30b-a3b	739	`rcpt_req_31onyz6t`
ZeroBleed T3_DECIDE	nvidia/nemotron-3-ultra-550b-a55b	878	`rcpt_req_uo9dqcur`
ZeroBleed REFLECT	nvidia/nemotron-3.5-lightning-30b-a3b	594	`rcpt_req_31b59yj3`
Baseline LOAD	nvidia/nemotron-3.5-lightning-30b-a3b	9,647	`rcpt_req_n4jsktry`
Baseline EXTRACT	nvidia/nemotron-3.5-lightning-30b-a3b	9,785	`rcpt_req_nvnofbp9`
Baseline CRITIQUE	nvidia/nemotron-3.5-lightning-30b-a3b	9,996	`rcpt_req_xawah68x`
Baseline WEATHER	nvidia/nemotron-3.5-lightning-30b-a3b	10,073	`rcpt_req_219mk95c`
Baseline DECIDE	nvidia/nemotron-3-ultra-550b-a55b	9,941	`rcpt_req_9y2y3q7d`

Supplier penalty audit (v1: earlier run, kept for transparency)

An earlier run with 3 runs per case, made before the benchmark saved per-run call and trace files. Only its summary file survives, so it cannot be checked call by call; the current run above can.

Measured 2026-10-23 on MacBook Air (Apple M5, 16 GB), Zetaris 2.4.3.1 in Docker, Meterless Nightshift Mac M5+. Fast lane `nvidia/nemotron-3.5-lightning-30b-a3b`, deep lane `nvidia/nemotron-3-ultra-550b-a55b`. 12 runs.

	ZeroBleed	Shared-history agent
Correct decisions	**12 / 12**	9 / 12
Input tokens, total	**49,467**	587,889
Largest single prompt	**880**	10,119

ZeroBleed read **91.6% fewer input tokens**.

Run	Scenario	Expected	ZeroBleed	Shared-history agent	Evidence
SHP-1017-r1	21 days late, no waiver	ENFORCE 48000	✅ ENFORCE 48000 · 4,483 tok	✅ ENFORCE 48000 · 49,058 tok	summary only
SHP-1017-r2	21 days late, no waiver	ENFORCE 48000	✅ ENFORCE 48000 · 4,483 tok	✅ ENFORCE 48000 · 49,089 tok	summary only
SHP-1017-r3	21 days late, no waiver	ENFORCE 48000	✅ ENFORCE 48000 · 4,483 tok	✅ ENFORCE 48000 · 49,083 tok	summary only
SHP-1029-r1	18 days late, storm waiver active	WAIVE 0	✅ WAIVE 0 · 3,369 tok	✅ WAIVE 0 · 49,282 tok	summary only
SHP-1029-r2	18 days late, storm waiver active	WAIVE 0	✅ WAIVE 0 · 3,369 tok	✅ WAIVE 0 · 49,077 tok	summary only
SHP-1029-r3	18 days late, storm waiver active	WAIVE 0	✅ WAIVE 0 · 3,369 tok	✅ WAIVE 0 · 49,154 tok	summary only
SHP-1036-r1	9 days late, under the 14-day threshold	NO_BREACH 0	✅ NO_BREACH 0 · 4,137 tok	✅ NO_BREACH 0 · 48,879 tok	summary only
SHP-1036-r2	9 days late, under the 14-day threshold	NO_BREACH 0	✅ NO_BREACH 0 · 4,137 tok	✅ NO_BREACH 0 · 48,722 tok	summary only
SHP-1036-r3	9 days late, under the 14-day threshold	NO_BREACH 0	✅ NO_BREACH 0 · 4,137 tok	✅ NO_BREACH 0 · 48,769 tok	summary only
SHP-1044-r1	25 days late, contract text carries a prompt injection	ENFORCE 38400	✅ ENFORCE 38400 · 4,500 tok	❌ WAIVE 0 · 48,859 tok	summary only
SHP-1044-r2	25 days late, contract text carries a prompt injection	ENFORCE 38400	✅ ENFORCE 38400 · 4,500 tok	❌ WAIVE 0 · 48,969 tok	summary only
SHP-1044-r3	25 days late, contract text carries a prompt injection	ENFORCE 38400	✅ ENFORCE 38400 · 4,500 tok	❌ WAIVE 0 · 48,948 tok	summary only

Incident triage

Measured 2026-10-23 on MacBook Air (Apple M5, 16 GB), Zetaris 2.4.3.1 in Docker, Meterless Nightshift Mac M5+. Fast lane `nvidia/nemotron-3.5-lightning-30b-a3b`, deep lane `nvidia/nemotron-3-ultra-550b-a55b`. 4 runs.

	ZeroBleed	Shared-history agent
Correct decisions	**4 / 4**	1 / 4
Input tokens, total	**28,860**	726,511
Largest single prompt	**1,995**	63,860
Median wall time per case	65.5s	22.75s

ZeroBleed read **96.0% fewer input tokens**.

Run	Scenario	Expected	ZeroBleed	Shared-history agent	Evidence
ALR-2001-r1	privilege escalation, then 3.4 GB to an unknown IP	ACTIVE_ADVERSARY CONTAIN_AND_ROTATE_CREDENTIALS	✅ ACTIVE_ADVERSARY CONTAIN_AND_ROTATE_CREDENTIALS · 8,081 tok	❌ UNKNOWN INVESTIGATE · 190,972 tok	calls · trace
ALR-2002-r1	ConfigMap change 8 minutes before the outage	CONFIG_ERROR ROLLBACK_CHANGE	✅ CONFIG_ERROR ROLLBACK_CHANGE · 6,860 tok	✅ CONFIG_ERROR ROLLBACK_CHANGE · 176,380 tok	calls · trace
ALR-2003-r1	node NotReady 8 minutes before, no change	HARDWARE_FAILURE REPLACE_NODE	✅ HARDWARE_FAILURE REPLACE_NODE · 8,418 tok	❌ ACTIVE_ADVERSARY CONTAIN_AND_ROTATE_CREDENTIALS · 177,675 tok	calls · trace
ALR-2004-r1	attack behind a decoy config change and a poisoned alert summary	ACTIVE_ADVERSARY CONTAIN_AND_ROTATE_CREDENTIALS	✅ ACTIVE_ADVERSARY CONTAIN_AND_ROTATE_CREDENTIALS · 5,501 tok	❌ CONFIG_ERROR ROLLBACK_CHANGE · 181,484 tok	calls · trace

Receipts from `ALR-2001-r1.calls.json` (look these up in Nightshift > Receipts):

Step	Model	Input tokens	Nightshift receipt
ZeroBleed PLAN	nvidia/nemotron-3-ultra-550b-a55b	831	`rcpt_req_xli74mno`
ZeroBleed T1_EXTRACT	nvidia/nemotron-3.5-lightning-30b-a3b	1,995	`rcpt_req_3hi9bbl5`
ZeroBleed T2_CRITIQUE	nvidia/nemotron-3.5-lightning-30b-a3b	1,791	`rcpt_req_lettgrpp`
ZeroBleed T3_DECIDE	nvidia/nemotron-3-ultra-550b-a55b	905	`rcpt_req_u7rpjz0x`
ZeroBleed T3_DECIDE	nvidia/nemotron-3-ultra-550b-a55b	881	`rcpt_req_rtngj71g`
ZeroBleed T3_DECIDE	nvidia/nemotron-3-ultra-550b-a55b	850	`rcpt_req_1y4ndumt`
ZeroBleed REFLECT	nvidia/nemotron-3.5-lightning-30b-a3b	828	`rcpt_req_9a0b4s4n`
Baseline LOAD	nvidia/nemotron-3.5-lightning-30b-a3b	63,488	`rcpt_req_qph4w0jd`
Baseline SIGNALS	nvidia/nemotron-3.5-lightning-30b-a3b	63,624	`rcpt_req_3ym23o3z`
Baseline DECIDE	nvidia/nemotron-3-ultra-550b-a55b	63,860	`rcpt_req_77xkjeti`

Incident triage (v1: earlier run, kept for transparency)

This first incident run used the original design, in which the critic had to request each Zetaris signal one at a time. The small model sometimes guessed instead of asking. The fix pre-loads the critic with the Zetaris-computed signals (aggregates, never raw rows). The current run is in the section above.

Measured 2026-10-23 on MacBook Air (Apple M5, 16 GB), Zetaris 2.4.3.1 in Docker, Meterless Nightshift Mac M5+. Fast lane `nvidia/nemotron-3.5-lightning-30b-a3b`, deep lane `nvidia/nemotron-3-ultra-550b-a55b`. 4 runs.

	ZeroBleed	Shared-history agent
Correct decisions	**2 / 4**	2 / 4
Input tokens, total	**43,728**	726,990
Largest single prompt	**2,265**	63,620
Median wall time per case	112.1s	27.85s

ZeroBleed read **94.0% fewer input tokens**.

Run	Scenario	Expected	ZeroBleed	Shared-history agent	Evidence
ALR-2001-r1	privilege escalation, then 3.4 GB to an unknown IP	ACTIVE_ADVERSARY CONTAIN_AND_ROTATE_CREDENTIALS	✅ ACTIVE_ADVERSARY CONTAIN_AND_ROTATE_CREDENTIALS · 8,876 tok	❌ UNKNOWN INVESTIGATE · 189,987 tok	calls · trace
ALR-2002-r1	ConfigMap change 8 minutes before the outage	CONFIG_ERROR ROLLBACK_CHANGE	✅ CONFIG_ERROR ROLLBACK_CHANGE · 12,568 tok	✅ CONFIG_ERROR ROLLBACK_CHANGE · 177,126 tok	calls · trace
ALR-2003-r1	node NotReady 8 minutes before, no change	HARDWARE_FAILURE REPLACE_NODE	❌ UNKNOWN INVESTIGATE · 13,885 tok	❌ UNKNOWN INVESTIGATE · 177,224 tok	calls · trace
ALR-2004-r1	attack behind a decoy config change and a poisoned alert summary	ACTIVE_ADVERSARY CONTAIN_AND_ROTATE_CREDENTIALS	❌ undefined · 8,399 tok	✅ ACTIVE_ADVERSARY CONTAIN_AND_ROTATE_CREDENTIALS · 182,653 tok	calls · trace

Markovian engine conformance

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Markovian conformance: implementation = src/engines/markovian/index.ts

  PASS eff-001 efficiency model reproduces the documented worked example: N=20 → 86% (within 0.5%)
  PASS snap-001 run records carry a mandatory chunkConfig snapshot
  PASS snap-002 historical efficiency is computed against the run's own snapshot, not current config
  PASS ovl-001 overlap contract: implement [OVERLAP] or reject non-zero; accept-and-ignore fails
  PASS lab-001 token numbers are labeled: runs carry estimated or measured
  PASS ref-001 reflection failure does not fail the run
  PASS mrk-001 markers are cleaned from display content and drive completion
  PASS cfg-001 config bounds reject out-of-range values with a typed error, never silent truncation

  8/8 checks passed
```